

AYMANE AIT DADS

Data Science Intern - Software Engineer | Computer Vision · Machine Learning · NDT Data Processing
Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and executing an end-to-end machine learning pipeline for a 30B-parameter model under GPU-constrained conditions. Developed computer vision models achieving 99.8% accuracy on aerial image classification and 0.791 AUC on AI-generated image detection, applying CLIP ViT-L/14, DenseNet121, and ensemble methods to extract actionable insights from high-volume image data. Brings hands-on experience in Python, PyTorch, Scikit-learn, and unsupervised anomaly detection directly applicable to processing inspection and NDT sensor data for industrial applications. Aims to apply computer vision and data correlation techniques to advance Screening Eagle's software products for infrastructure inspection and predictive maintenance.

CORE COMPETENCIES

Computer Vision | Machine Learning | NDT Data Processing | Anomaly Detection | Data Correlation | Feature Engineering | Predictive Maintenance | Inspection Data Analysis

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means models on 100,000+ fiber-optic network records** - including upload speed, latency, and jitter - to classify network quality across thousands of nodes.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, data preprocessing, and systematic model evaluation workflows built in Python and Scikit-learn.
- Delivered stakeholder presentations translating **machine learning model outputs** into actionable maintenance recommendations, bridging technical findings and business decision-making using Power BI.

PROJECTS

AI-Generated Image Detection - NTIRE EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in EURECOM Class 2026 @ CVPR

- Ranked **1st in the EURECOM class private Kaggle leaderboard (0.791 AUC)** detecting AI-generated images across 250,000 training samples and 25+ generator types; independently submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Fine-tuned **CLIP ViT-L/14 (428M parameters)** with a custom MLP head and dual learning rates, applying 10-view TTA and a 3-model ensemble weighted by validation AUC to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80

- Designed an **unsupervised fault detection system for slide-rail industrial machinery** achieving AUC 0.80 for predictive maintenance, using a Transformer autoencoder trained on Mel spectrograms with SpecAugment augmentation.
- Engineered the full pipeline - from raw audio to **anomaly scoring without labeled anomaly examples** - directly mirroring the challenge of uncovering hidden data correlations in NDT inspection sensor streams.

PyTorch · Mel Spectrograms · Transformer Autoencoder · SpecAugment

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle · June 2026 · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) on structured reasoning tasks.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified CoT data.

PyTorch · Unsloth · PEFT · TRL · vLLM · Triton · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Computer Vision: PyTorch | Scikit-learn | OpenCLIP | CLIP ViT | DenseNet | CNN | TensorFlow

Data Science & Engineering: Python | Pandas | NumPy | Feature Engineering | Clustering | Ensemble Methods | Power BI

LLM Fine-Tuning & MLOps: LoRA | PEFT | Unsloth | Hugging Face Transformers | BF16/FP16 | Git | Linux | Docker